

8.6 Consider the wage equation

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + e_i \quad (XR8.6a)$$

where wage is measured in dollars per hour, education and experience are in years, and $METRO = 1$ if the person lives in a metropolitan area. We have $N = 1000$ observations from 2013.

- We are curious whether holding education, experience, and $METRO$ constant, there is the same amount of random variation in wages for males and females. Suppose $\text{var}(e_i | \mathbf{x}_i, FEMALE = 0) = \sigma_M^2$ and $\text{var}(e_i | \mathbf{x}_i, FEMALE = 1) = \sigma_F^2$. We specifically wish to test the null hypothesis $\sigma_M^2 = \sigma_F^2$ against $\sigma_M^2 \neq \sigma_F^2$. Using 577 observations on males, we obtain the sum of squared OLS residuals, $SSE_M = 97161.9174$. The regression using data on females yields $\hat{\sigma}_F^2 = 12.024$. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.
- We hypothesize that married individuals, relying on spousal support, can seek wider employment types and hence holding all else equal should have more variable wages. Suppose $\text{var}(e_i | \mathbf{x}_i, MARRIED = 0) = \sigma_{SINGLE}^2$ and $\text{var}(e_i | \mathbf{x}_i, MARRIED = 1) = \sigma_{MARRIED}^2$. Specify the null hypothesis $\sigma_{SINGLE}^2 = \sigma_{MARRIED}^2$ versus the alternative hypothesis $\sigma_{MARRIED}^2 > \sigma_{SINGLE}^2$. We add $FEMALE$ to the wage equation as an explanatory variable, so that

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + \beta_5 FEMALE + e_i \quad (XR8.6b)$$

Using $N = 400$ observations on single individuals, OLS estimation of (XR8.6b) yields a sum of squared residuals is 56231.0382. For the 600 married individuals, the sum of squared errors is 100,703.0471. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.

- Following the regression in part (b), we carry out the NR² test using the right-hand-side variables in (XR8.6b) as candidates related to the heteroskedasticity. The value of this statistic is 59.03. What do we conclude about heteroskedasticity, at the 5% level? Does this provide evidence about the issue discussed in part (b), whether the error variation is different for married and unmarried individuals? Explain.
- Following the regression in part (b) we carry out the White test for heteroskedasticity. The value of the test statistic is 78.82. What are the degrees of freedom of the test statistic? What is the 5% critical value for the test? What do you conclude?
- The OLS fitted model from part (b), with usual and robust standard errors, is

$$\begin{array}{cccccc} \widehat{WAGE} & = & -17.77 & + & 2.50EDUC & + & 0.23EXPER & + & 3.23METRO & - & 4.20FEMALE \\ (se) & & (2.36) & (0.14) & (0.031) & & (1.05) & & (0.81) \\ (robse) & & (2.50) & (0.16) & (0.029) & & (0.84) & & (0.80) \end{array}$$

For which coefficients have interval estimates gotten narrower? For which coefficients have interval estimates gotten wider? Is there an inconsistency in the results?

- If we add $MARRIED$ to the model in part (b), we find that its t -value using a White heteroskedasticity robust standard error is about 1.0. Does this conflict with, or is it compatible with, the result in (b) concerning heteroskedasticity? Explain.

$$(a) \begin{cases} H_0: \sigma_M^2 = \sigma_F^2 \\ H_1: \sigma_M^2 \neq \sigma_F^2 \end{cases}, N_m = 577$$

$$\hat{\sigma}_M^2 = \frac{97161.9174}{577-4} = 169.567$$

$$\hat{\sigma}_F^2 = 12.024^2 = 144.5766$$

$$GR = \frac{\hat{\sigma}_M^2}{\hat{\sigma}_F^2} = \frac{169.567}{144.5766} \approx 1.173$$

$$F_{0.025}(573, 1000-577-4) = F_{0.025}(573, 419) \approx (0.838, 1.197)$$

$0.838 < 1.173 < 1.197 \rightarrow$ fail to reject H_0 , we don't have evidence to show that $\hat{\sigma}_M^2 \neq \hat{\sigma}_F^2$ at 95% level of significance.

$$\hat{\sigma}_M^2 = SSE_M / df_M$$

$$F = \hat{\sigma}_M^2 / \hat{\sigma}_F^2$$

$$(b) \begin{cases} H_0: \sigma_s^2 = \sigma_m^2 \\ H_1: \sigma_m^2 > \sigma_s^2 \end{cases}$$

$$\hat{\sigma}_s^2 = \frac{56231.0382}{400-5} \approx 142.36$$

$$\hat{\sigma}_m^2 = \frac{100703.0471}{600-5} \approx 169.25$$

$$\hat{F}_{0.05} = \frac{169.25}{142.36} \approx 1.189$$

$F_{0.05}(595, 395) \approx 1.165 < 1.189 \rightarrow \text{reject } H_0$, we have evidence to show that $\hat{\sigma}_m^2 > \hat{\sigma}_s^2$.

(c)

$$NR^2 = 59.03, K=5$$

$$\chi^2 = N \times R^2 \sim \chi^2_{(S-1)}$$

$$59.03 \sim \chi^2_{(5-1)} = \chi^2(4)$$

$\chi^2(0.95, 4) \approx 9.488 < NR^2 = 59.03 \rightarrow \text{reject } H_0$, we have evidence to show that the existing of heteroskedasticity at 95% significant level.

The NR^2 test evaluates whether the variance of errors between observations in the regression model is constant, but it does not specifically examine whether the error variation differs between married and unmarried individuals. Therefore, although the result of the NR^2 test suggests potential heteroskedasticity, it does not directly provide evidence regarding the specific issue discussed in the question.

(d)

虛擬變數 METRO 和 FEMALE 的平方等於其本身，故不重複計算。

EDUC, EXPER, METRO, FEMALE,

EDUC², EXPER²,

EDUC × EXPER, EDUC × METRO, EDUC × FEMALE, EXPER × METRO,

EXPER × FEMALE, METRO × FEMALE

$$d.f = 4 + 2 + 6 = 12, \alpha = 0.05$$

$$\chi^2_{0.95, 12} \approx 21.026 < \chi^{2*} = 78.82 \rightarrow \text{reject } H_0 \text{ at } 5\% \text{ critical value.}$$

We have significant evidence to reject homoskedasticity. It means $\sigma_{SINGLE}^2 \neq \sigma_{MARRIED}^2$

(e)

Since robust standard error is bigger than usual standard errors for intercept and EDUC, the interval estimates for the intercept and the coefficient of EDUC have gotten wider.

Because robust standard error is smaller than usual standard errors for the rest variables, so the interval estimates for the others have gotten narrower.

From the significance of indicator results, we can find that it is consistent between two standard errors.

(f)

The addition of the intercept indicator variable *MARRIED* allows the expected wage, conditional on the explanatory variables *EDUC*, *EXPER*, *METRO*, and *FEMALE* to differ between those who are married and those who are not.

In part (b), we were asking about the amount of error variation between the two groups (married & single), which we found to be significant.

So the above two issues are quite different.

compatible

∴ (b) 做了異質性 test 且存在異質性，
而 (f) 用了 robust se 來估計不顯著，
∴ 合理，

8.16 A sample of 200 Chicago households was taken to investigate how far American households tend to travel when they take a vacation. Consider the model

$$MILES = \beta_1 + \beta_2 INCOME + \beta_3 AGE + \beta_4 KIDS + e$$

MILES is miles driven per year, *INCOME* is measured in \$1000 units, *AGE* is the average age of the adult members of the household, and *KIDS* is the number of children.

- Use the data file *vacation* to estimate the model by OLS. Construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant.
- Plot the OLS residuals versus *INCOME* and *AGE*. Do you observe any patterns suggesting that heteroskedasticity is present?
- Sort the data according to increasing magnitude of income. Estimate the model using the first 90 observations and again using the last 90 observations. Carry out the Goldfeld–Quandt test for heteroskedastic errors at the 5% level. State the null and alternative hypotheses.
- Estimate the model by OLS using heteroskedasticity robust standard errors. Construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant. How does this interval estimate compare to the one in (a)?
- Obtain GLS estimates assuming $\sigma_i^2 = \sigma^2 INCOME_i^2$. Using both conventional GLS and robust GLS standard errors, construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant. How do these interval estimates compare to the ones in (a) and (d)?

(a)

```
Call:
lm(formula = miles ~ income + age + kids, data = vacation)
```

Residuals:

Min	1Q	Median	3Q	Max
-1198.14	-295.31	17.98	287.54	1549.41

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-391.548	169.775	-2.306	0.0221	*
income	14.201	1.800	7.889	2.10e-13	***
age	15.741	3.757	4.189	4.23e-05	***
kids	-81.826	27.130	-3.016	0.0029	**

Signif. codes:

0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

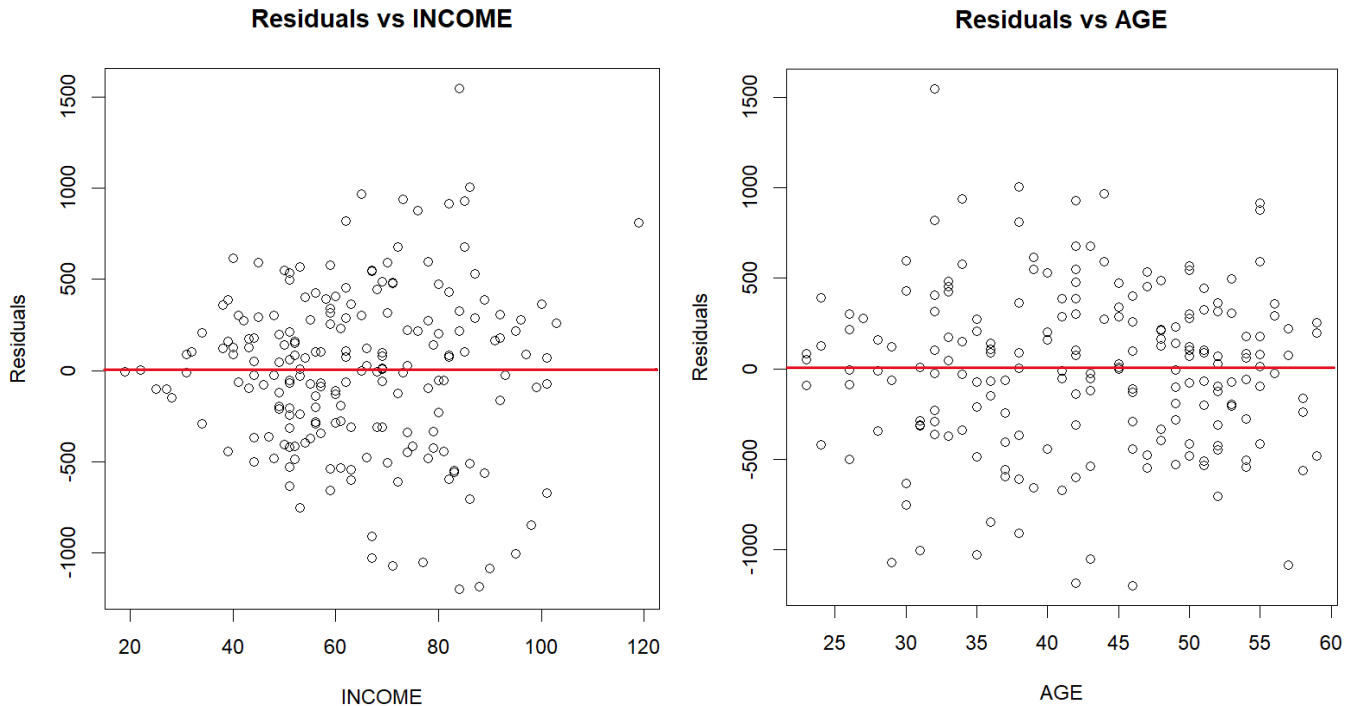
Residual standard error: 452.3 on 196 degrees of freedom
Multiple R-squared: 0.3406, Adjusted R-squared: 0.3305
F-statistic: 33.75 on 3 and 196 DF, p-value: < 2.2e-16

```
>
> confint(ols_model, "kids", level = 0.95)
      2.5 %      97.5 %
kids -135.3298 -28.32302
```

Since the interval does not include zero, the effect of an additional child on miles traveled is statistically significant and negative, holding income and age constant.

(b)

income $\uparrow$, residuals $\uparrow \Rightarrow$ 可能存在異質變異性



As Income increases, the dispersion of the residuals becomes larger. This provides strong visual evidence of heteroskedasticity suggesting that the variance of the error term increases with income.

The residuals appear more tightly clustered at younger ages and more dispersed in middle age ranges. While not a perfect funnel shape, this suggests possible mild heteroskedasticity with respect to age.

Overall, both plots are broadly consistent with the presence of heteroskedasticity, supporting the result of formal tests. This implies that OLS standard errors may be inconsistent, and therefore heteroskedasticity-robust standard errors or GLS-type methods would be more appropriate for inference.

(c)

$$\begin{cases} H_0: \sigma_{low}^2 = \sigma_{high}^2 \\ H_1: \sigma_{low}^2 \neq \sigma_{high}^2 \end{cases}$$

```
> cat("GQ statistic:", round(gq_stat, 3), "\n")
GQ statistic: 3.104
> cat("Critical value at 5% level:", round(f_crit, 3), "\n")
Critical value at 5% level: 1.429
```

The computed F-statistic is 3.1041, while the 5% critical value is 1.4286. Since the test statistic is greater than the critical value, we reject the null hypothesis.

This provides evidence of heteroskedasticity, indicating that the variance of the error term differs between low-income and high-income households.

(d)

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-391.5480	142.6548	-2.7447	0.0066190
income	14.2013	1.9389	7.3246	6.083e-12
age	15.7409	3.9657	3.9692	0.0001011
kids	-81.8264	29.1544	-2.8067	0.0055112

(Intercept) **
income ***
age ***
kids **

Signif. codes:

0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
> cat("\n95% robust confidence interval for 'kids':\n")
```

95% robust confidence interval for 'kids':

```
> cat("(", round(lower, 4), ", ", round(upper, 4), ")\n")  
( -138.969 , -24.6838 )
```

The 95% confidence interval using heteroskedasticity-robust standard errors is (-138.969, -24.6838).

Compared with part (a), the interval is slightly different due to the use of robust standard errors.

However, both intervals remain entirely below zeros, indicating that the effect of an additional child on miles traveled is significantly negative in both cases. Therefore, the qualitative conclusion remains unchanged.

(e)

```
> cat("GLS (conventional SE) 95% CI for kids:\n")  
GLS (conventional SE) 95% CI for kids:  
> cat("(", round(ci_gls_conventional[1], 4), ", ", round(ci_gls_conventional[2], 4), ")\n\n")  
( -119.8945 , -33.7181 )
```

```
>  
> vcov_gls_robust <- vcovHC(gls_model, type = "HC1")  
> coef_kids_gls <- coef(gls_model)["kids"]  
> se_kids_gls_robust <- sqrt(vcov_gls_robust["kids", "kids"])  
>  
> lower_gls_robust <- coef_kids_gls - 1.96 * se_kids_gls_robust  
> upper_gls_robust <- coef_kids_gls + 1.96 * se_kids_gls_robust  
>  
> cat("GLS (robust SE) 95% CI for kids:\n")  
GLS (robust SE) 95% CI for kids:  
> cat("(", round(lower_gls_robust, 4), ", ", round(upper_gls_robust, 4), ")\n")  
( -121.1388 , -32.4738 )
```

Compared with parts (a) and (d), the GLS-based intervals has narrower confidence interval.

All intervals (OLS, robust OLS, and GLS) lead to the same qualitative conclusion, an additional child significantly reduces miles traveled, as all confidence intervals remain entirely below zero.